Group: Ctrl+S

OpsResolve

Domain 1 : AI System & Agentic Workflows Automation →

Github link:

HTTPS://GITHUB.COM/AOAQ1121/OPSRESOLVE

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Problem Statement

Fragmented Manual Workflows

- Departments use disconnected systems
- Heavy reliance on human coordination
- No centralized platform

Unstructured Communication

- WhatsApp, emails, spreadsheets
- No formal record of decisions
- Information lost in chat threads

No Real-Time Synchronization

- Information delays between departments
- Cannot see conflicts as they happen
- Decisions based on outdated data

Repeated Operational Clashes

- Same conflicts happen again and again
- No learning from past issues
- Wasted time and resources

SWOT Analysis

Strengths

- AI-powered detection
- 5-min resolution time
- Full audit trail
- Role-based access
- Recurring conflict tracking

Weakness

- Requires internet
- Manual role selection
- No real WebSocket yet
- Limited to 3 departments

Opportunities

- ERP integration (SAP)
- Mobile app development
- Multi-plant deployment
- Predictive analytics

Threats

- Competitor solutions
- AI API costs
- Data privacy regulations
- Enterprise adoption barriers

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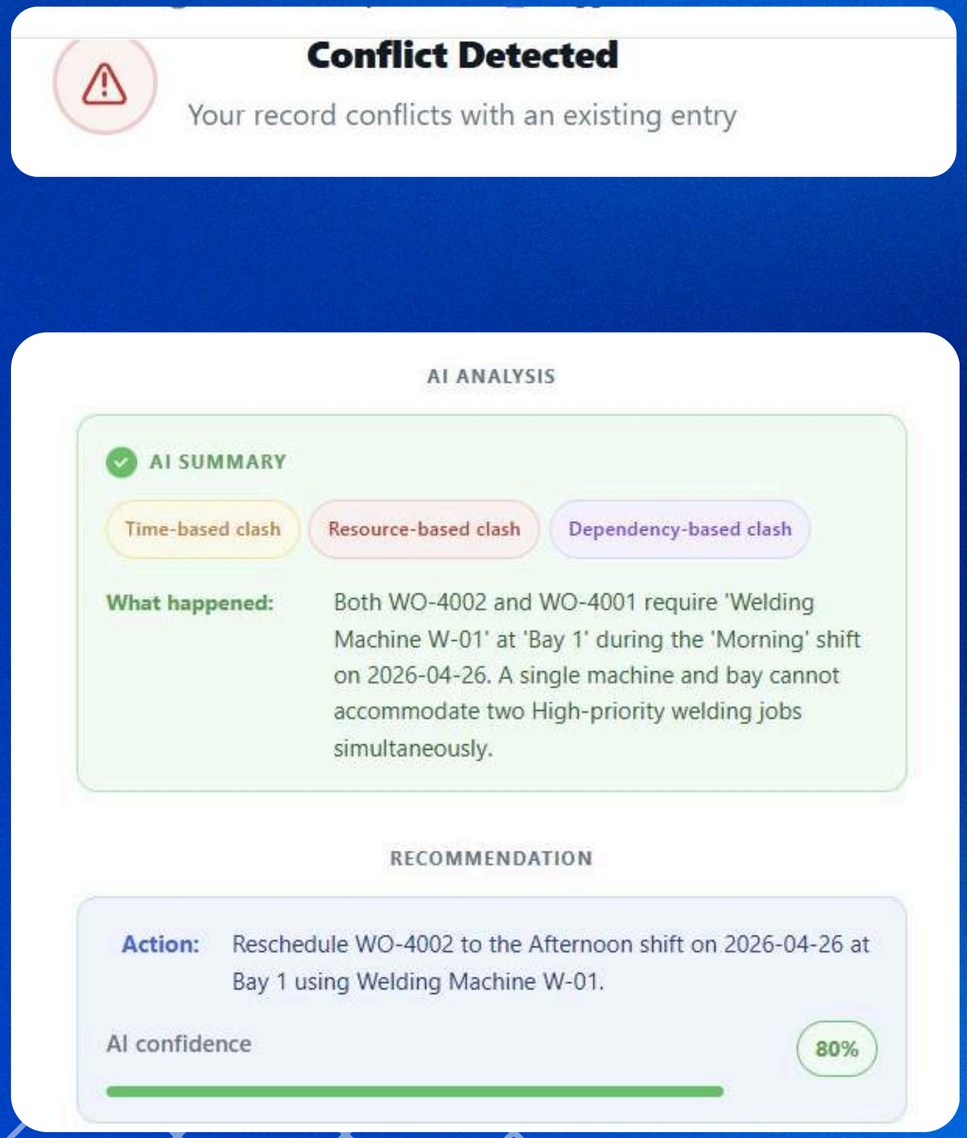
Solution Overview

- Detects operational conflicts automatically
- Understands unstructured input using Gemini 2.5 Flash
- Performs multi-step reasoning
- Orchestrates workflow decisions dynamically



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Key Features

Real-Time Conflict Detection

Instant alert on submit - catches cross-department clashes the moment a record is added

AI Processing

4-agent pipeline: Parser - Detector - Decision Engine - Explanation

AI Fill & Imputation

Upload a PDF - Gemini extracts all fields.

Conflict Management

Department-scoped view with severity, AI confidence score and full conflict analysis

Machine Request

Say or type your need - AI extracts machine, shift and date, then show available options

Decision Module

Role based authority - notify, schedule, accept AI recommendation or override

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LIVE DEMO

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Technical Architecture

Frontend
React + Vite

Backend
Node.js + Express

AI
Gemini 2.5 Flash

Database
Firebase Firestore

State
Custom StateManager

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Multi-Agent AI System

Input Understanding

- "Maintenance tomorrow but production needs machine urgently"
- Gemini extracts: resource, departments, time, urgency

Conflict Detection

- Detects schedule overlap, resource clashes, budget conflicts

Cause + Impact Analysis

- Gemini analyzes why conflict happened
- Assesses severity (High/Medium/Low)
- Predicts operational impact

Decision & Recommendation

- Evaluates options and chooses best solution

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Status WorkFlow

SUBMITTED → UNDER ANALYSIS → CONFLICT DETECTED → UNDER REVIEW → RESOLVED

Future Roadmap

Phase 2: WebSocket notifications, Analytics dashboard, Mobile app

Phase 3: SAP/Oracle integration, Slack/Teams bot, Predictive forecasting

Phase 4 : Multi-plant deployment, Custom AI fine-tuning

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Thank You

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